

Graph Theory

Conceptual Questions & Answers
Discrete Mathematics | Lectures 12 · 13 · 14 · 16 · 17

Lecture 12 — Trees — Definitions & Properties

Q1. [conceptual]

Can two different rooted trees on the same set of vertices be considered the same unrooted tree? Give an example to support your answer.

Answer

Yes. When you root an unrooted tree at different vertices, the parent-child relationships change completely — children become parents and vice versa — even though the underlying edge set is identical.

- Example: a path graph 1—2—3. Rooted at 1, node 2 is the child of 1 and node 3 is the child of 2. Rooted at 2, nodes 1 and 3 are both children of 2. These are structurally different rooted trees, yet the unrooted graph is the same.

This shows that the unrooted tree is the more 'fundamental' object; a rooted tree carries extra information (the choice of root) that is not intrinsic to the graph itself.

Q2. [conceptual]

A student claims: 'If I remove any single edge from a tree, I get two smaller trees.' Is this always true? What exactly are those two pieces?

Answer

Yes, always. Every edge in a tree is a bridge — removing it disconnects the graph into exactly two components.

- Each component is connected (by definition of connected component) and acyclic (subgraphs of acyclic graphs are acyclic). A connected acyclic graph is a tree.
- If the original tree has n vertices and the removed edge separates it into components of size k and $n-k$, then one component is a tree on k vertices (with $k-1$ edges) and the other is a tree on $n-k$ vertices (with $n-k-1$ edges).

This 'divide into two trees' property is unique to trees — removing an edge from a graph with a cycle may leave it still connected.

Q3. [conceptual]

What is the minimum and maximum number of leaves a tree on n vertices ($n \geq 2$) can have? Describe the tree shape that achieves each extreme.

Answer

Minimum leaves = 2 | Maximum leaves = $n - 1$

- Minimum (2 leaves): a path graph — a straight chain $v_1-v_2-v_3-\dots-v_n$. Every internal vertex has degree 2; only the two endpoints have degree 1 (are leaves). A path is the 'most linear' tree possible.
- Maximum ($n-1$ leaves): a star graph — one central vertex connected to all $n-1$ other vertices. Every non-central vertex is a leaf. This is the 'most branchy' tree possible.

Any tree with $n \geq 2$ vertices falls somewhere between these two extremes. The number of leaves is closely tied to how many high-degree vertices exist: each vertex of degree d 'contributes' to d branches, supporting more leaves.

Q4. [conceptual]

In a full binary tree, can two internal vertices be adjacent (connected by an edge) while both having degree 3? What does this tell you about the structure of internal vertices?

Answer

Yes, absolutely. Two internal vertices can share an edge, and both can have degree 3 (two children and one parent, except the root which has degree 2).

This is actually the norm in a full binary tree — internal vertices form a connected subtree among themselves. If you remove all leaves from a full binary tree, what remains is still a connected subtree (a tree of internal vertices).

- The root is the only internal vertex with degree 2 (no parent, two children). All other internal vertices have degree 3 (one parent, two children).
- This means the internal vertices are never isolated from each other — they form the 'skeleton' of the tree, with leaves hanging off at the boundary.

Q5. [conceptual]

A complete binary tree has all leaves at the same level. How does adding just one more leaf (making it no longer complete) affect the height of the tree?

Answer

If the complete binary tree already has all 2^h leaves at level h (a perfect binary tree), adding one more leaf means opening a new level $h+1$. The height increases by 1.

However, if the tree is complete but not perfect — meaning the last level is not fully filled — you can add a leaf at the current last level (from left to right) without increasing the height at all.

Height only increases when the last level is already completely full before the insertion.

This is why complete binary trees are used in heap data structures: insertions are cheap ($O(\log n)$) because the height grows very slowly — only after every level is full.

Q6. [conceptual]

If you know a rooted binary tree has height h , what can you say about the minimum and maximum number of vertices it can contain?

Answer

Minimum vertices = $h + 1$ | Maximum vertices = $2^{(h+1)} - 1$

- Minimum: a path — each level has exactly one vertex. A tree of height h with one vertex per level has $h+1$ vertices total.
- Maximum: a perfect binary tree — every level is completely filled. Level k has 2^k vertices, so total = $1 + 2 + 4 + \dots + 2^h = 2^{(h+1)} - 1$.

This wide range is important: two trees can have the same height but vastly different numbers of vertices. Height alone doesn't tell you much about size — only about the longest root-to-leaf path.

Lecture 13 — Three Characterisations of Trees

Q7. [conceptual]

A connected graph G has n vertices and n edges. What can you say about G ? Is it a tree? Does it contain a cycle?

Answer

G is NOT a tree. A tree on n vertices has exactly $n-1$ edges. G has one extra edge beyond the tree threshold.

A connected graph on n vertices with n edges contains exactly one cycle.

Intuition: start with a spanning tree ($n-1$ edges, no cycles). Adding one more edge between any two vertices that already have a path between them creates exactly one new cycle. The extra edge, together with the existing tree path between its endpoints, forms a unique cycle.

- Such a graph is sometimes called a unicyclic graph — it has exactly one cycle, with trees hanging off the cycle vertices as 'branches'.

Q8. [conceptual]

Why is every edge in a tree a 'bridge' (cut edge)? What would it mean structurally if a tree had even one non-bridge edge?

Answer

In a tree, removing any edge disconnects the graph (by definition — trees are minimally connected). A bridge is exactly an edge whose removal disconnects the graph. So every tree edge is a bridge.

If a tree had a non-bridge edge e , then removing e would leave the graph still connected. That means there is another path between e 's endpoints — but then e together with that path forms a cycle. A tree with a cycle is a contradiction.

Tree = connected + every edge is a bridge = connected + acyclic.

This is why trees feel 'fragile' compared to general graphs — there is no redundancy. Every edge is load-bearing.

Q9. [conceptual]

Two graphs G and H both have 7 vertices and 6 edges. G is connected; H is not. Describe what H must look like. Can H contain a cycle?

Answer

H has 7 vertices, 6 edges, but is disconnected. Since a connected graph on 7 vertices needs at least 6 edges (a spanning tree), H is 'just barely' disconnected — it has as many edges as a tree but split across components.

- H must have at least 2 connected components. Each component is itself a connected graph. The components together account for all 7 vertices and 6 edges.
- If one component has k vertices and is a tree, it uses $k-1$ edges. The remaining $7-k$ vertices share $6-(k-1) = 7-k$ edges. For the remaining piece to also be a tree it needs $(7-k)-1 = 6-k$ edges. But $7-k \neq 6-k$, so at least one component must have a cycle.

Yes — H can contain a cycle. For example: one component is a 6-cycle (6 vertices, 6 edges) and one isolated vertex. Connected? No. Cycle? Yes.

Q10. [conceptual]

A graph is called a 'forest' if it is acyclic (but not necessarily connected). What is the relationship between the number of edges, vertices, and connected components in a forest?

Answer

In a forest, each connected component is itself a tree (connected + acyclic). So the edge-count rule for trees applies to each component separately.

For a forest with n vertices, m edges, and k connected components: $m = n - k$

- If $k = 1$ (the forest is connected), this gives $m = n-1$, recovering the tree formula.
- If $k = n$ (every vertex is isolated, no edges), then $m = n-n = 0$. Correct — a graph with no edges and n vertices has n components.

This formula is extremely useful: if you know any two of $\{m, n, k\}$ for a forest, you immediately know the third. It also means adding an edge to a forest either reduces k by 1 (connecting two components) or creates a cycle (if both endpoints are already in the same component).

Q11. [conceptual]

If a graph G has n vertices, $n-1$ edges, but is disconnected, what can you say about G ? Can it still be acyclic?

Answer

Yes, G can be acyclic. A disconnected acyclic graph is a forest. A forest with n vertices and k components has $n-k$ edges. If G has $n-1$ edges and is a forest, then $n-k = n-1$, giving $k = 1$ — but $k = 1$ means connected. Contradiction.

A disconnected graph on n vertices with $n-1$ edges cannot be acyclic — it must contain at least one cycle.

Here is why: if it were acyclic, it would be a forest. A forest with $k \geq 2$ components has at most $n-2$ edges (since $m = n-k \leq n-2$). But G has $n-1$ edges — too many for a disconnected forest. The extra edge must create a cycle within one of the components.

- Example: a triangle (3 vertices, 3 edges, one component, one cycle) plus an isolated vertex = 4 vertices, 3 edges = $n-1$ edges, disconnected, and contains a cycle.

Lecture 14 — Prüfer Codes & Labeled Trees

Q12. [conceptual]

Two different labeled trees on the same vertex set always produce different Prüfer sequences. What does this tell you about the information content of the Prüfer sequence — does it fully capture the tree's structure?

Answer

Yes — the Prüfer sequence is a complete encoding of a labeled tree. Because distinct trees give distinct sequences (the encoding is injective) and every sequence decodes to a unique tree (the decoding is surjective), the Prüfer sequence captures every structural detail of the labeled tree.

- This means vertex labels, adjacency, and degree information are all embedded in the sequence. Knowing the Prüfer sequence is exactly as good as knowing the tree itself.
- In contrast, the Prüfer sequence of an unlabeled tree would be meaningless — the encoding depends entirely on the specific numerical labels assigned to vertices.

Prüfer sequence $\leftrightarrow$ labeled tree is a perfect bijection. No information is lost or added.

Q13. [conceptual]

If every entry in a Prüfer sequence is the same number — say the sequence is [5, 5, 5, 5, 5, 5] for a tree on 8 vertices — what does the tree look like?

Answer

When one vertex appears in every position of the Prüfer sequence, that vertex is the neighbour of every removed leaf. It is the last vertex standing (other than vertex n) throughout the algorithm.

The tree is a star — vertex 5 is the central hub connected to all other 7 vertices.

- Degree of vertex 5 = (times it appears in sequence) + 1 = 6 + 1 = 7. It is connected to all other vertices.
- All other vertices appear 0 times in the sequence, so their degree = 0 + 1 = 1. They are all leaves.

This is a clean example of reading tree structure directly off the sequence: a constant sequence always encodes a star graph centred at that repeated vertex.

Q14. [conceptual]

The Prüfer sequence for a path graph 1—2—3—4—5 (in that linear order) is [2, 3, 4]. What pattern do you notice, and will this pattern always hold for any path graph on n vertices?

Answer

In a path 1—2—3—...— n , vertex 1 is the smallest leaf, removed first (its neighbour 2 is recorded). Then vertex 2 becomes a leaf, removed next (neighbour 3 recorded). This continues until two vertices remain.

Prüfer sequence of the path 1—2—3—...— n is always [2, 3, 4, ..., $n-1$].

- The sequence lists the 'interior' vertices in order, skipping the two endpoints (1 and n).
- Every vertex in the sequence appears exactly once, confirming that path vertices have degree 2 (appear once $\rightarrow$ degree $1+1 = 2$), while the endpoints have degree 1 (appear 0 times $\rightarrow$ degree $0+1 = 1$).

This is a satisfying sanity check: the most 'linear' tree produces the most 'sequential' Prüfer sequence.

Q15. [conceptual]

Cayley's formula says there are n^{n-2} labeled trees on n vertices. For $n = 3$ this gives 3 trees, and for $n = 4$ it gives 16. Can you verify the $n = 3$ case by simply listing all labeled trees on vertices $\{1, 2, 3\}$?

Answer

A tree on 3 vertices has exactly 2 edges. The three possible trees on $\{1, 2, 3\}$ are:

- Path with 2 as the centre: edges $\{1, 2\}$ and $\{2, 3\}$. Vertex 2 has degree 2; vertices 1 and 3 are leaves.
- Path with 1 as the centre: edges $\{1, 2\}$ and $\{1, 3\}$. Vertex 1 has degree 2; vertices 2 and 3 are leaves.
- Path with 3 as the centre: edges $\{1, 3\}$ and $\{2, 3\}$. Vertex 3 has degree 2; vertices 1 and 2 are leaves.

Total = 3 = $3^{3-2} = 3^1 = 3$. ✓

All three trees are structurally identical (all are paths on 3 vertices) but they are distinct as labeled trees because the vertex playing the 'centre' role differs. Labeling makes them different objects.

Q16. [conceptual]

A Prüfer sequence has length $n-2$. What are the first and last entries of the sequence when the tree is a path $1-2-3-\dots-n$? What vertex can NEVER appear as the last entry in any Prüfer sequence, and why?

Answer

For the path $1-2-3-\dots-n$ the sequence is $[2, 3, \dots, n-1]$. The first entry is 2 and the last entry is $n-1$.

Vertex n can never be the last entry (or any entry) in a Prüfer sequence.

Here is why: the last entry in the sequence is the neighbour of the second-to-last leaf removed. At every step, we remove the smallest-labeled current leaf. Vertex n , having the highest label, is never the smallest leaf at any step — it can never be removed during the algorithm. It always survives to be one of the final two vertices. Since it is never removed, its neighbour is never recorded, so n never appears anywhere in the sequence.

- Similarly, the two vertices remaining at the very end (whose connecting edge is the 'free' final edge) always include vertex n .

Q17. [conceptual]

A graph is drawn with edge crossings. Your friend says 'this graph is non-planar.' Are they necessarily right? How would you correctly argue the point?

Answer

No — they are wrong. A graph is non-planar only if no crossing-free drawing exists. A single drawing with crossings proves nothing about planarity.

Planarity is a property of the abstract graph, not of any particular drawing.

- Counter-example: K_4 is planar, but it is easy to accidentally draw it with a crossing. The existence of one bad drawing does not make it non-planar.
- To prove a graph IS planar, you just need to exhibit one crossing-free drawing.
- To prove a graph is NOT planar, you must show that no crossing-free drawing can possibly exist — this requires a theorem like Kuratowski's or the edge-count inequality.

This is a common misconception. Always ask: does there EXIST a crossing-free drawing? Not: is this particular drawing crossing-free?

Q18. [conceptual]

Euler's formula gives $n - m + f = 2$ for connected plane graphs. What does the formula become if the graph is disconnected with k connected components? Explain intuitively why the formula changes.

Answer

For a plane graph with k connected components: $n - m + f = k + 1$

Intuition: each time you disconnect a graph into one more component, you are essentially 'separating' the plane but not merging any faces. Adding an isolated vertex (increasing k by 1) adds 1 to n without changing m or f , so $n-m+f$ increases by 1.

- For $k=1$ (connected): $n-m+f = 2$. This is the standard Euler formula.
- For $k=2$: $n-m+f = 3$. The outer face is shared, but the components are 'independent islands' in the plane.

Another way to see it: take a connected plane graph (satisfying $n-m+f=2$) and remove a bridge. This splits it into 2 components; n stays the same, m drops by 1, f stays the same. So $n-m+f$ becomes 3. Now $k=2$ and the formula $k+1=3$ holds.

Q19. [conceptual]

What is the minimum number of faces any connected plane graph can have, regardless of how many vertices and edges it has? What kind of graph achieves this minimum?

Answer

Minimum faces = 1 (just the outer/unbounded face).

A connected plane graph that is also a tree has no cycles. With no cycles, it is impossible to enclose any bounded region — so the only face is the outer (unbounded) face.

- Verify with Euler's formula: a tree has $m = n - 1$ edges. So $f = 2 - n + m = 2 - n + (n - 1) = 1$. ✓

As soon as you add one edge to a tree (creating a cycle), you enclose a bounded region and f jumps to 2. Every additional cycle edge adds exactly one new face. So the face count directly counts 'how many extra edges beyond a tree' the graph has.

Q20. [conceptual]

In a plane graph, is it possible for a single edge to be on the boundary of the same face on both sides? When does this happen?

Answer

Yes — this happens exactly when the edge is a bridge (a cut edge). Removing a bridge disconnects the graph, and the 'face' on both sides of the bridge is actually the same face, just reached by going around the bridge in two directions.

- Example: a tree edge. In any plane drawing of a tree, every edge is a bridge, and every edge borders the single outer face on both of its sides.
- For a non-bridge edge (one that lies on a cycle), the edge separates two distinct faces — one on each side.

An edge borders the same face on both sides ■ it is a bridge.

This is why the face-counting formula $2m \geq 3f$ counts bridge edges twice toward the same face: it is a natural consequence of the bridge structure.

Q21. [conceptual]

Kuratowski's theorem uses the concept of a 'subdivision.' Why is it necessary to allow subdivisions — why not just say a graph is non-planar if and only if it contains K_5 or $K_{3,3}$ as a subgraph?

Answer

Because inserting degree-2 vertices along edges does not affect planarity. If you can draw H without crossings, you can draw any subdivision of H without crossings (just route the path along the same curve as the original edge). So planarity is preserved by subdivision.

Many non-planar graphs do not contain K_5 or $K_{3,3}$ as direct subgraphs but do contain them as subdivisions. For example, the Petersen graph does not literally contain $K_{3,3}$ as a subgraph (it has only 10 vertices and each has degree 3), but it contains a subdivision of $K_{3,3}$.

Subdivision captures 'topological equivalence': graphs that are homeomorphic to K_5 or $K_{3,3}$ are just as non-planar as the originals.

- If we required exact subgraphs, Kuratowski's theorem would miss infinitely many non-planar graphs.

Lecture 17 — Maximal Planar Graphs & Degree Bounds

Q22. [conceptual]

What is the difference between a maximal planar graph and a maximum planar graph? Is every maximal planar graph also a maximum planar graph?

Answer

These sound similar but mean different things:

- Maximal planar graph: a planar graph to which no edge can be added while keeping it planar. It is locally 'full' — you cannot improve it without violating planarity.
- Maximum planar graph: a planar graph with the largest possible number of edges among all planar graphs on n vertices. This is the globally optimal graph.

Every maximum planar graph is maximal, but not every maximal planar graph is maximum.

Counter-example: consider a planar graph where one face is a quadrilateral (4-cycle). You could add a diagonal to triangulate it, so the graph is not maximal. But if that quadrilateral face cannot be triangulated without crossing another edge (due to the graph's specific embedding), it might be maximal in that embedding even without being maximum overall.

However, for simple graphs on $n \geq 3$ vertices, all maximal planar graphs (triangulations) have exactly $3n-6$ edges, so they are all maximum.

Q23. [conceptual]

The inequality $m \leq 3n - 6$ gives an upper bound on edges in a planar graph. For which specific values of n is this bound actually achieved? What do those graphs look like?

Answer

The bound $m = 3n - 6$ is achieved by triangulated plane graphs (maximal planar graphs) — plane graphs where every face, including the outer face, is a triangle.

For $n \geq 3$, the bound is always achievable. For $n = 3$: $m = 3(3) - 6 = 3$, which is K_3 (a triangle). For $n = 4$: $m = 6$, which is K_4 .

- $n=3$: K_3 — a single triangle, 3 edges. Every face (two of them, including outer) is a triangle. ✓
- $n=4$: K_4 — 4 vertices, 6 edges. Draw a triangle with a vertex inside connected to all three corners. Every face is a triangle. ✓
- $n=5$: $3(5) - 6 = 9$ edges. One example is K_5 minus one edge — but wait, that has 9 edges and is planar (K_5 itself is not planar, but removing one edge makes it so).

For larger n , triangulations can always be constructed by starting with a triangle and repeatedly inserting a vertex inside a face and connecting it to all three boundary vertices.

Q24. [conceptual]

The theorem 'every planar graph has a vertex of degree at most 5' is used in proving the four-colour theorem. Intuitively, why does planarity limit how 'connected' any single vertex can be?

Answer

In the plane, if a vertex has very high degree, many edges radiate from it like spokes. Those edges divide the surrounding area into many narrow 'wedges.' Each wedge must contain a face. As the degree grows, the faces get squeezed into a tighter and tighter ring around the vertex.

But Euler's formula tightly constrains how many faces a planar graph can have ($f = 2 - n + m \leq 2 - n + 3n - 6 = 2n - 4$). There simply aren't enough faces to 'support' a vertex of very high degree without violating the global edge bound.

High degree in a planar graph → many edges → too many edges → violates $m \leq 3n - 6$. So high degree is forbidden.

- Degree 6 or more for every vertex would require $m \geq 3n$ edges (by handshaking), contradicting $m \leq 3n - 6$.

The threshold is exactly 5: the icosahedron is a planar graph where every vertex has degree exactly 5, and it achieves 5-regularity without any contradiction.

Q25. [conceptual]

For bipartite planar graphs, the bound tightens to $m \leq 2n - 4$ instead of $m \leq 3n - 6$. Intuitively, why does being bipartite make the graph 'less dense' as a planar graph?

Answer

In a bipartite graph, vertices are split into two groups and edges only go between groups. This means no odd-length cycles exist — in particular, no triangles (3-cycles).

In a planar triangulation, every face is a triangle (3 edges on boundary). This is what allows the maximum density $m = 3n - 6$. But in a bipartite graph, triangular faces are impossible — every face must be bounded by at least 4 edges.

No triangles → every face needs at least 4 boundary edges → fewer faces can 'fit' → fewer edges overall.

- Analogy: imagine tiling a room with polygons. If you are forbidden from using triangles and must use squares or larger, you can tile the same room with fewer polygon-boundaries. Fewer boundary segments = fewer edges in the graph.

This is why $K_{3,3}$ (the complete bipartite graph on 3+3 vertices) is non-planar even though $9 \leq 3(6) - 6 = 12$ passes the general test: the tighter bipartite bound $2(6) - 4 = 8$ catches it.

Q26. [conceptual]

A planar graph is said to be 'outerplanar' if it can be drawn in the plane with all vertices on the outer face boundary. What is the maximum number of edges an outerplanar graph on n vertices can have?

Answer

In an outerplanar graph, all vertices lie on the boundary of the outer face. Every interior face is bounded by edges that connect vertices along the outer boundary — and since no vertex is 'inside', every interior face must be a triangle (you can always triangulate a polygon).

Maximum edges in an outerplanar graph on n vertices = $2n - 3$.

- Intuition: start with a cycle on all n vertices (n edges). Then triangulate the interior of the cycle by adding non-crossing diagonals. A convex polygon on n vertices can be triangulated with $n - 3$ diagonals, giving $n + (n - 3) = 2n - 3$ edges total.
- Compare: planar bound is $3n - 6$, bipartite planar is $2n - 4$, outerplanar is $2n - 3$. Outerplanar is even more restrictive because all vertices must lie on the outer face.

K_4 is planar but not outerplanar: you cannot place all 4 vertices of K_4 on the outer face without a crossing.

Q27. [conceptual]

Can a planar graph be regular (every vertex has the same degree)? If yes, which regular degrees are possible, and can you give an example for each?

Answer

Yes. The degree bound (at least one vertex has degree ≤ 5) does not prevent regularity — it just caps the minimum degree. A planar graph can be k -regular for $k = 0, 1, 2, 3, 4$, or 5 .

- 0-regular: isolated vertices (trivially planar).
- 1-regular: a perfect matching — a set of disjoint edges (trivially planar).
- 2-regular: a disjoint union of cycles — always planar.
- 3-regular (cubic): the cube graph Q_3 (8 vertices, 12 edges). Also the Petersen graph is 3-regular but NOT planar — so 3-regular does not guarantee planarity.
- 4-regular: the octahedron graph (6 vertices, 12 edges) is 4-regular and planar.
- 5-regular: the icosahedron graph (12 vertices, 30 edges) is 5-regular and planar. Check: $m = 30 \leq 3(12) - 6 = 30$. ✓ It hits the bound exactly.

6-regular planar graphs do NOT exist: 6-regularity would give $m \geq 3n$, contradicting $m \leq 3n - 6$.